

Amplitude-Independent Waveform Descriptors of Earthquake Size: A Leakage-Controlled Gradient-Boosting Study on Turkish Broadband Data

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Abstract

Most data-driven magnitude estimators exploit recorded amplitude, which is the quantity magnitude scales are defined from. This study asks how much information about earthquake size remains in a seismogram once absolute amplitude, and every quantity that acts as an amplitude proxy, is removed, and which physical measurement carries it. From 78 732 three-component broadband records of 11 758 earthquakes ($M \geq 2.0$) recorded by the KOERI network in 2025–2026, 131 descriptors were extracted, each constructed to be invariant to a rescaling of the record and none referenced to the pre-event noise. A within-event audit removed 79 descriptors whose values tracked station signal-to-noise ratio. Gradient-boosted regression trees were trained on a magnitude-balanced, spatially de-clustered subset of 3 697 events and evaluated on folds that share neither events nor stations with the training data. The amplitude-independent model reached an event-level mean absolute error of 0.343 ± 0.017 magnitude units, compared with 0.392 ± 0.016 for source–station geometry alone and 0.238 ± 0.022 for a reference model given peak amplitude, i.e. it recovered about one third of the gap between the two. Permutation tests identify the network-median, attenuation-corrected corner frequency of P and S waves as the dominant descriptor, ahead of source–station distance. Its scaling with magnitude, however, is much shallower than self-similar source theory predicts, indicating a band-limited apparent corner frequency; correspondingly, all models underestimate events above M 3.5. These results are preliminary.

1 Introduction

Earthquake magnitude scales are defined from amplitudes: local magnitude M_L from the peak of a simulated Wood–Anderson record, moment magnitude M_w from the low-frequency level of the displacement spectrum (Hanks and Kanamori, 1979). Machine-learning estimators trained on raw waveforms inherit this definition and learn primarily from amplitude, either explicitly or through normalisation choices that preserve it (Mousavi and Beroza, 2020, 2023). Such estimators are accurate, but they answer a narrow question and are exposed to the weaknesses of amplitude: instrument gain and site amplification errors, clipping near large events, and saturation of early, short windows.

Source physics predicts that earthquake size also controls the *shape* of a seismogram. Larger ruptures last longer and radiate relatively more low-frequency energy, which lowers the corner frequency of the source spectrum (Brune, 1970; Boatwright, 1980; Abercrombie, 1995). Period parameters used in earthquake early warning exploit the same principle from the first seconds of the P wave (Allen and Kanamori, 2003; Kanamori, 2005; Wu and Kanamori, 2005). These shape descriptors are, however, entangled with propagation (attenuation, distance) and with the noise floor of weak records, so a data-driven model can easily appear to learn “shape” while in fact recovering amplitude.

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This study therefore asks two questions: **(i)** how much information about magnitude is retained by waveform descriptors that are provably insensitive to amplitude and empirically insensitive to signal-to-noise ratio, and **(ii)** which physical descriptor carries that information. The contributions are:

- a feature set of 131 descriptors, each invariant to a rescaling of the record by construction and verified by an automated test, including physics-based source, period and duration parameters;
- a within-event leakage audit that separates noise-floor sensitivity from genuine source variation, and is applied by default;
- a magnitude-balanced, spatially de-clustered event selection and a doubly disjoint (event and station) validation design with explicit geometric and amplitude reference models;
- a ranking of descriptors by grouped permutation importance, which identifies the corner frequency, corrected for attenuation, as the main carrier of size information.

2 Background and literature review

2.1 Magnitude scales and source scaling

Moment magnitude is tied to the static seismic moment (Hanks and Kanamori, 1979). Local magnitude, by contrast, is a dynamic, high-frequency measure; Deichmann (2006) showed theoretically that the two scales diverge for small earthquakes, whose corner frequencies lie above the passband of the Wood–Anderson response. Since the AFAD catalogue reports M_L for most events in this study and M_w for larger ones, the target variable is a mixture of the two scales. In the ω^2 source model (Brune, 1970) the displacement spectrum is flat below the corner frequency f_c and decays as f^{-2} above it. Under constant stress drop (self-similarity), $M_0 \propto f_c^{-3}$, i.e. $\log_{10} f_c$ decreases by 0.5 per unit M_w . Self-similarity of small earthquakes has been supported by borehole recordings down to very small magnitudes (Abercrombie, 1995) and by stacked P and S spectra of southern California earthquakes (Prieto et al., 2004), although the question remains debated.

2.2 Spectral source parameters and attenuation

Observed spectra combine source, path and site. High-frequency decay near the receiver is commonly modelled by the parameter κ (Anderson and Hough, 1984); for the Marmara region, Sertçelik et al. (2022) reported a dependence of κ on magnitude, which illustrates how strongly source and attenuation effects are coupled. Parametric source fits include the Brune model and the generalised fall-off model of Boatwright (1980). A non-parametric alternative replaces the fitted corner by the ratio of spectral integrals of velocity and displacement (Andrews, 1986; Snoke, 1987), which is more stable for band-limited data. Frequency-dependent attenuation in Anatolia is strong: $Q_S \approx 40f^{1.03}$ was reported for the Sea of Marmara (Horasan and Boztepe-Güney, 2004), and coda $Q_c \approx 57.5f^{0.82}$ for the East Anatolian Fault Zone (Sertçelik, 2012). At magnitudes 2–3 the expected corner frequencies (of order 10 Hz or more) fall in the band where such attenuation removes most energy over typical regional distances.

2.3 Period and duration parameters from early warning

Earthquake early warning estimates size from the first seconds of P. The recursive predominant period $\tau_p^{\max}$ (Allen and Kanamori, 2003) and the effective period τ_c (Kanamori, 2005; Wu and Kanamori, 2005) are ratios of signal moments and therefore independent of amplitude; Olson and Allen (2005) argued from such observations that final size is partly determined early in the

rupture. Colombelli and Zollo (2015) showed that the logarithm of peak P displacement as a function of time (LPDT) grows and then levels off, and related the corner time of this curve to the rupture half-duration. Significant duration, the interval over which normalised cumulative energy grows from 5 to 95 % (Trifunac and Brady, 1975), is likewise amplitude-free, and modern duration models parameterise its magnitude dependence through source duration (Afshari and Stewart, 2016).

2.4 Machine-learning magnitude estimation

Deep networks trained on single-station waveforms estimate local magnitude with small error (Mousavi and Beroza, 2020; Lomax et al., 2019), and multi-station transformer models have been developed for early warning (Münchmeyer et al., 2021); see Mousavi and Beroza (2023) for a review. These models are designed to use amplitude and are not intended to separate it from other information. Gradient-boosted decision trees (Ke et al., 2017) on engineered features offer a transparent alternative, for which exact tree-based Shapley values (Lundberg et al., 2020) and permutation importance (Breiman, 2001) are available; for correlated features, conditional or grouped schemes are preferable (Strobl et al., 2008).

2.5 Leakage and structured validation

Leakage, information about the target entering through an unintended route, inflates apparent skill (Kaufman et al., 2011), and models readily exploit such shortcuts (Geirhos et al., 2020). For data with spatial and hierarchical structure, random cross-validation is optimistic and blocked designs are recommended (Roberts et al., 2017). In this problem, amplitude can leak through signal-to-noise-dependent shapes, station selection and catalogue metadata, and site response can leak across folds that share stations.

2.6 Gap addressed

To the author’s knowledge, the literature reviewed here does not quantify how much magnitude information remains when amplitude and its proxies are removed by construction and audited empirically, nor ranks physically interpretable descriptors under a validation design that excludes shared events and shared stations. A systematic literature search with full-text access is required to confirm this assessment.

3 Data

Hypocentres and magnitudes were taken from the AFAD earthquake catalogue (AFAD, 2026). Waveforms are continuous 100 Hz three-component broadband (HH) recordings of the KOERI network (KOERI, 1971), cut from 60 s before to 120 s after each catalogue origin time, together with station coordinates. Windows were available for 12 617 catalogue events with $M \geq 2.0$ between May 2025 and August 2026. Magnitudes are reported to 0.1 units (12 257 M_L , 360 M_w) and origin times to whole seconds. The magnitude distribution is strongly skewed: the median is 2.3 and about 70 % of events lie between 2 and 3 (Figure 1).

4 Methodology

4.1 Pre-processing, travel times and P picking

For each event and station the first instrument with three components was selected (HH in all retained records). Segments were merged, and any gap inside the required time span rejected the record. Theoretical first P and S arrivals were interpolated from a table of the iasp91 model

(Kennett and Engdahl, 1991) computed with TauP (Crotwell et al., 1999) as implemented in ObsPy (Beyreuther et al., 2010). The required span was demeaned, linearly detrended, tapered (5 % cosine) and filtered with causal fourth-order Butterworth filters (1–40 Hz band-pass for time-domain descriptors and picking; 0.5 Hz high-pass for spectra and period parameters). Instrument response was not removed.

The P onset was refined with the Akaike-information-criterion picker of Maeda (1985) on the band-passed vertical component, searching from 1.5 s before to 3 s after the theoretical arrival (and at least 0.3 s before the theoretical S); the window is asymmetric because catalogue origin times are given to whole seconds. The S time was set to the picked P time plus the theoretical S–P interval.

4.2 Windows and quality control

Fixed windows relative to the picks were used throughout (Table 1), so that spectral resolution and statistics are comparable between records. A record was rejected if it was clipped (absolute count $\geq 8 \times 10^6$, or at least five samples at the peak value), or if the root-mean-square signal-to-noise ratio (SNR) in the P or S window, relative to a noise window ending 1 s before P, was below 3. SNR was used only for this gate and for the audit below, never as a model input.

Table 1: Analysis windows (t_P : picked P time; t_S : derived S time).

Window	Span	Components
Noise (QC only)	$[t_P - 6, t_P - 1)$ s	all
P	$[t_P - 0.05, t_P + 1.95)$ s (200 samples)	Z
S	$[t_S - 0.2, t_S + 4.8)$ s (500 samples)	N, E
Polarisation	first 1 s of the P and S windows	Z, N, E
Envelope	$[t_P - 1, t_S + 10)$ s	horizontal
LPDT	$[t_P, t_P + \min(3, t_S - t_P - 0.2))$ s	Z

4.3 Amplitude invariance and the leakage audit

Every descriptor is computed after normalising each window by its own peak, and is either a ratio of quantities of equal degree in amplitude, a time or frequency read from a normalised curve, an ordinal statistic, or a fitted shape whose level parameter is discarded. Consequently, multiplying a record by any constant $c \neq 0$ leaves every descriptor unchanged, which is verified by an automated test on synthetic records.

Invariance does not exclude indirect leakage: a fixed-length window on a weak record contains relatively more noise, so spectral and envelope shapes can encode SNR, and thereby amplitude. Partial correlation with SNR at fixed catalogue magnitude is an unsuitable test, because catalogue magnitudes are uncertain and a genuinely larger event has both higher SNR and a lower corner frequency. The audit therefore operates *within events*, where the source is identical. Ranks of a descriptor x , of $\log_{10}$ SNR and of $\log_{10}$ hypocentral distance r are demeaned per event; x and SNR are then residualised on r , and the leakage score is

$$\lambda(x) = \max_{w \in \{P, S\}} \left| \text{corr} \left(\tilde{x}, \widetilde{\log \text{SNR}_w} \right) \right|, \quad (1)$$

the larger absolute within-event partial rank correlation with the P- and S-window SNR. Descriptors with $\lambda > 0.15$, or with an undefined score, are excluded by default; the audit is recomputed on each training set. The threshold removed 79 of 131 descriptors (Table 2); the retained ones have a median λ of 0.07. Only results obtained with this filter are reported.

4.4 Descriptors

Table 2 summarises the families; all 131 are defined, with formulas, in the project documentation (`docs/FEATURES.md`). The key definitions are as follows.

Spectra. Velocity power spectra $P(f)$ were estimated with the multitaper method (Thomson, 1982) (time–bandwidth 2.5, four tapers) on the peak-normalised, high-passed P (Z) and S (N, E averaged) windows, in the bands 1–40 Hz and 0.7–40 Hz. Shape descriptors comprise the log-frequency centroid, entropy, peak frequency, roll-off frequencies and octave-band energy fractions.

Attenuation correction. Before source estimation the power spectrum was corrected for anelastic attenuation along the path,

$$P_{qc}(f) = P(f) \exp\left(\frac{2\pi f t}{Q(f)}\right), \quad Q(f) = Q_0 f^n, \quad (2)$$

with t the travel time, $Q_0^S = 60$, $n = 0.85$ and $Q_0^P = 1.5 Q_0^S$, values within the range reported for Anatolia (Horasan and Boztepe-Güney, 2004; Sertçelik, 2012). Geometric spreading is frequency independent and cancels in the shape.

Corner frequency. The non-parametric estimate of Andrews (1986) and Snoke (1987) is

$$f_c = \frac{1}{2\pi} \left(\frac{\int P_{qc}(f) df}{\int P_{qc}(f) (2\pi f)^{-2} df} \right)^{1/2}, \quad (3)$$

the square root of the ratio of velocity to displacement spectral energy. For an ω^2 spectrum over a wide band it returns f_c ; over a finite band it is an apparent, band-limited estimate. Brune and Boatwright model fits were also computed; only their residual misfits passed the audit.

LPDT. Following Colombelli and Zollo (2015), with u the drift-corrected displacement of the high-passed P window, $y(t) = \log_{10} (\max_{t' \leq t} |u| / \max_{t' \leq T} |u|)$ was fitted by a ramp-and-plateau function $a + b \min(t, T_{pl})$; the plateau time T_{pl} , the initial slope b and the time to 90 % of the final peak are retained.

Other descriptors. Peak-ratio periods $2\pi \text{PGV/PGA}$; polarisation from the covariance eigen-decomposition (rectilinearity, planarity, incidence, $\log H/V$); horizontal envelope shape (rise time, time to peak relative to S–P, decay exponent); and $|\text{skewness}|$ and crest factor.

Event medians and controls. Single-station estimates are affected by radiation pattern, directivity and site. For four key descriptors (P and S corner frequency, T_{pl} , and the P/S corner ratio) the median over all records of the same event *within the same training or test subset* was added. The number of stations was not used, because it tracks magnitude through recording selection. Epicentral and hypocentral distance, depth, back-azimuth and theoretical S–P time were supplied as controls, so that distance effects on shape are not attributed to the source.

4.5 Magnitude-balanced event selection

Events with at least three accepted records were binned in 0.2-unit magnitude bins, each containing exactly two catalogue values. Each bin was capped at 500 events; smaller bins were kept entirely, and no event was duplicated. Within each bin, events were grouped into space–time cells ($1^\circ \times 1^\circ \times \text{calendar year}$) and drawn round-robin over cells in random order, so that swarms

Table 2: Descriptor families: extracted and retained after the leakage audit.

Family	Extracted	Retained
Spectral shape, P and S (centroid, entropy, roll-off, band fractions, ...)	30	17
Source fits, uncorrected and attenuation-corrected (Brune, Boatwright, Snoke f_c)	32	14
P/S corner-frequency ratios	2	2
Early-warning periods (τ_c , $\tau_p^{\max}$)	2	0
Peak-ratio periods	4	2
LPDT (plateau time, slope, t_{90} , misfit)	4	3
Significant duration (D5–75, D5–95, D20–80, onset)	8	0
Waveform statistics (kurtosis, skew , crest, Hjorth, ...)	12	4
Complexity (permutation, sample, SVD entropy; Higuchi dimension)	8	0
Polarisation and S/P energy ratio	9	6
Envelope shape, Z and horizontal	20	4
Total record-level descriptors	131	52
+ event medians / controls		4 / 5

and aftershock sequences cannot dominate a bin. Finally, no cell was allowed to contribute more than 20 % of any bin (minimum three events), including bins below the cap. Without this limit, the 2025 Sındırgı (Balıkesir) sequence supplied 24–52 % of every bin above M 2.8 but about 1 % below it, a regional signature that a model could use in place of source information and that a station-disjoint split does not remove.

4.6 Model and validation

Gradient-boosted regression trees (Ke et al., 2017) were trained with a Huber loss and early stopping on an inner event-disjoint validation split (hyperparameters in the repository). Record weights were set so that each event’s records sum to one.

Events and stations were each partitioned into five groups. Fold k tests on records whose event *and* station belong to group k and trains on records whose event and station belong to neither, so that no event and no site response is shared. The partition was repeated three times with different random draws, giving 15 folds; metrics are reported as mean $\pm$ standard deviation over folds. The event prediction is the median of its station predictions. Four feature sets were trained on identical folds: *shape* (retained descriptors, event medians, controls), *shape only* (without controls), *controls only*, and an *amplitude reference* (logarithm of peak band-passed counts plus controls), which approximates a fitted local-magnitude relation and serves as an upper reference.

4.7 Importance analysis

Mean absolute Shapley values were computed with the exact tree algorithm (Lundberg et al., 2020). Because descriptors are correlated, permutation importance (Breiman, 2001) was computed for groups shuffled with one shared row permutation (cf. Strobl et al., 2008): (i) per *quantity*, joining a station-level descriptor with its event median; and (ii) per *concept*, joining substitutable quantities (e.g. P and S corner frequency). Importance is the increase in event-level mean absolute error (MAE), averaged over three shuffles and 15 folds.

5 Results

5.1 Dataset

Of 136 263 station records, 78 732 from 11 758 events and 151 stations passed quality control; the main rejection causes were low P-window SNR (49 583), gaps (3 761) and low S-window SNR (2 670). Of the 9 871 events with at least three records, selection retained 3 697 (3 451 M_L , 246 M_w) with 31 476 records, a median of seven records per event and a median hypocentral distance of 113 km (interquartile range 75–144 km). Bins up to M 2.6 were capped at 500 events; from M 2.8 upward the 20 % cell limit was binding instead (Figure 1). Owing to the doubly disjoint design, 3 561 events were tested at least once.

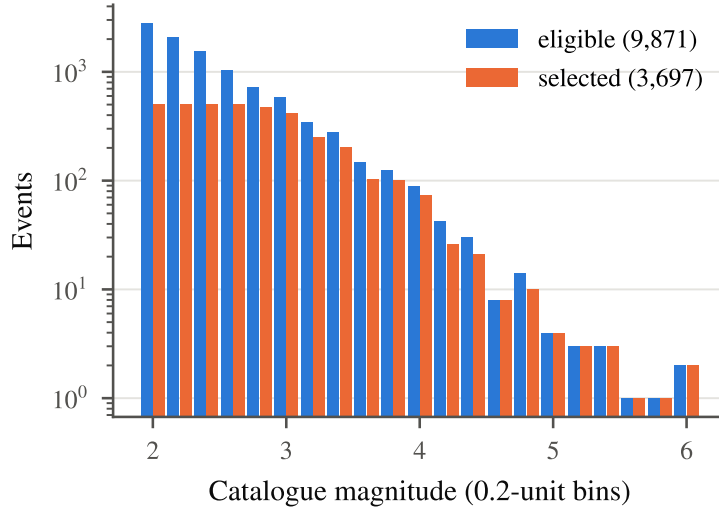


Figure 1: Magnitude distribution of events with at least three accepted records (eligible) and of the balanced selection (logarithmic scale).

5.2 Predictive skill

Table 3 compares the feature sets. Source–station geometry alone reduces the error relative to the constant predictor, reflecting that larger events are recorded at larger distances. The amplitude-independent descriptors reduce the event-level MAE from 0.392 to 0.343, which is 32 % of the difference between the geometry-only and amplitude reference models. Removing the controls changes the shape model only marginally. Figure 2 shows the event-level predictions.

Table 3: Event-level skill (mean $\pm$ s.d. over 15 doubly disjoint folds, $\approx$ 564 test events per fold). Record-level MAE in the last column.

Feature set	MAE	RMSE	R^2	Spearman ρ	MAE (records)
Constant (training mean)	0.466	–	–	–	–
Controls only	0.392 ± 0.016	0.525	0.21	0.49	0.446
Shape only	0.347 ± 0.018	0.452	0.42	0.63	0.369
Shape + controls	0.343 ± 0.017	0.446	0.43	0.64	0.365
Amplitude reference	0.238 ± 0.022	0.305	0.73	0.84	0.279

5.3 Magnitude-dependent bias

All models regress towards the mean of the training distribution (Table 4). Events at M 2.0–2.5 are overestimated by 0.2–0.4 units, partly because no smaller events exist in the catalogue to

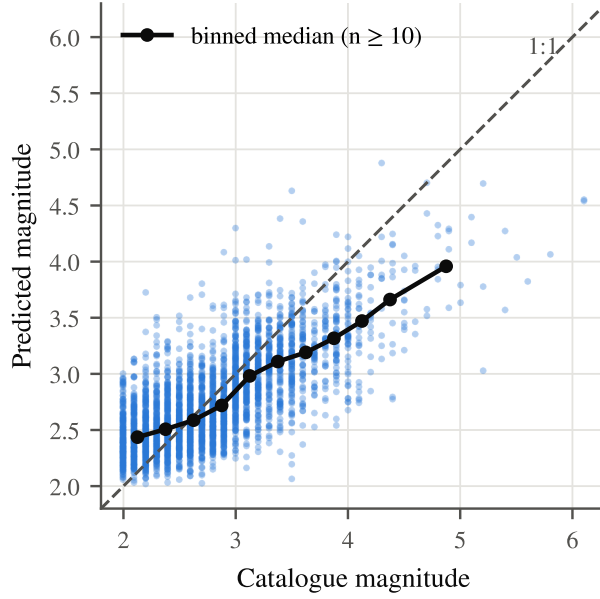


Figure 2: Predicted versus catalogue magnitude for the shape model (event median over stations, averaged over repeats; 3561 events).

balance them. Above M 3.5 the shape model underestimates increasingly, by -0.67 at M 4.0–4.5 and -1.01 above M 4.5. This is less than the geometry-only model but more than the amplitude reference. Near the centre of the distribution (M 2.5–3.0) the geometry-only model attains a lower MAE than the shape model, which is consistent with its predictions concentrating near the mean.

Table 4: Event-level bias (predicted – catalogue) and MAE by magnitude bin.

Bin	n	Shape + controls		Controls only		Amplitude ref.	
		bias	MAE	bias	MAE	bias	MAE
2.0–2.5	1186	+0.29	0.31	+0.38	0.38	+0.20	0.22
2.5–3.0	1161	−0.07	0.22	−0.09	0.15	−0.03	0.17
3.0–3.5	757	−0.16	0.28	−0.30	0.34	−0.08	0.19
3.5–4.0	306	−0.44	0.48	−0.73	0.73	−0.26	0.29
4.0–4.5	109	−0.67	0.69	−1.17	1.17	−0.39	0.40
≥ 4.5	42	−1.01	1.01	−1.89	1.89	−0.42	0.45

5.4 Descriptor importance

In the Shapley analysis, the event-median attenuation-corrected corner frequency of the P wave had the largest mean absolute contribution (0.18 magnitude units), followed by epicentral distance (0.12) and the corresponding S-wave corner frequency (0.12). Permutation of single quantities (Figure 3b) ranks epicentral distance first ($\Delta\text{MAE } 0.039 \pm 0.009$) and the P corner frequency second (0.020 ± 0.008 , positive in all 15 folds), followed by the misfit of the attenuation-corrected S-wave source fit, the S corner frequency, and the LPDT plateau time.

Single-quantity permutation understates correlated quantities that can substitute for each other. Shuffled jointly (Figure 3a), the P and S corner frequencies increase the MAE by 0.086 ± 0.012 , exceeding all distance controls together (0.073 ± 0.015); adding the remaining corner estimates and P/S ratios does not increase this value (0.084). The LPDT measures together contribute 0.008 ± 0.004 . All four concept effects were positive in every fold. Because permutation

substitutes misleading values rather than removing a feature, these increases measure the model’s reliance on a descriptor, not the loss expected after retraining without it.

The source-fit misfit has almost no correlation with magnitude by itself (partial Spearman $\rho = 0.01$ at fixed distance). Its importance therefore probably reflects an interaction: the misfit informs the model about the reliability of the corner-frequency estimate on a given record.

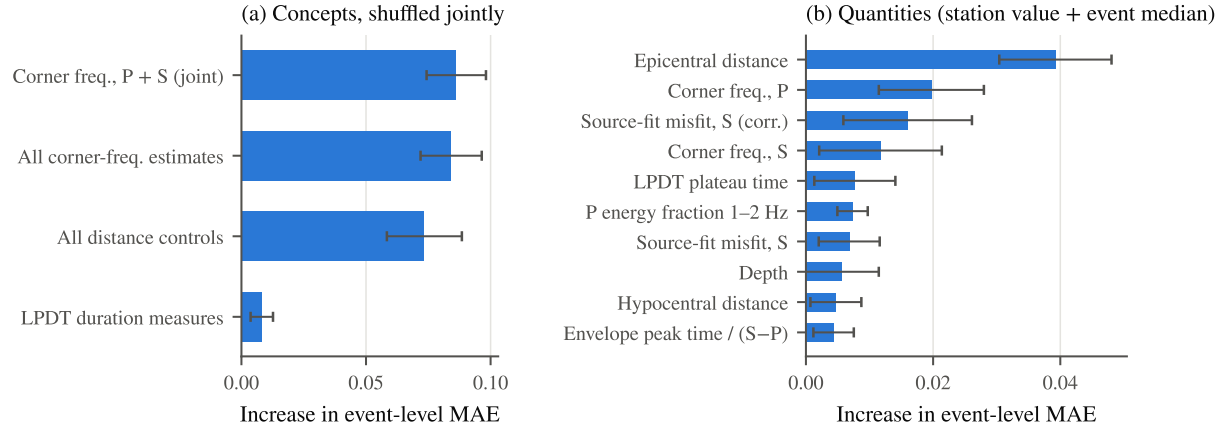


Figure 3: Grouped permutation importance (increase in event-level MAE; mean $\pm$ s.d. over 15 folds). (a) Substitutable quantities shuffled jointly. (b) Ten most important quantities, each shuffled together with its event median.

5.5 Corner-frequency scaling

The event-median apparent corner frequency decreases with magnitude for both phases (Figure 4; Spearman $\rho = -0.55$ for P and -0.49 for S). The fitted slopes of $\log_{10} f_c$ against magnitude are -0.13 (P) and -0.11 (S), and the medians lie between about 2 and 5 Hz. Self-similar scaling implies -0.5 per unit M_w ; even allowing for the faster growth of M_L relative to M_w at small magnitudes (Deichmann, 2006), the observed slope is several times shallower. Moreover, corner frequencies of a few hertz are low for M 2–3 earthquakes. The estimate is therefore an apparent, band-limited quantity that retains the ordering of source size but compresses its range.

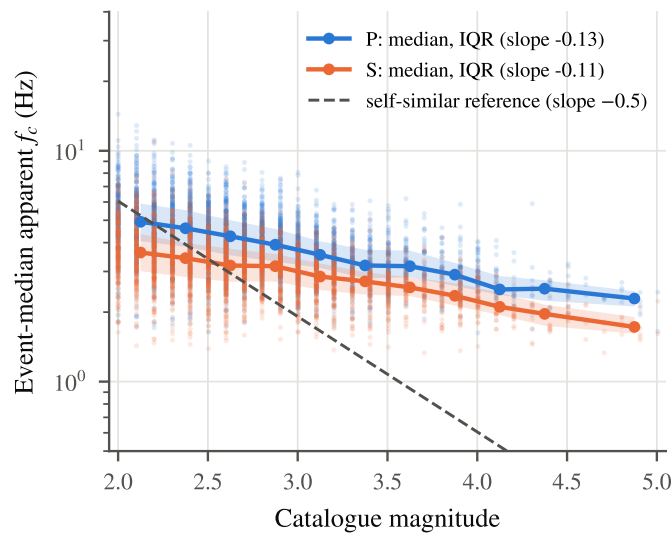


Figure 4: Event-median apparent corner frequency (Eq. 3) against magnitude; points are events, lines and bands are bin medians and interquartile ranges. The dashed line shows the self-similar slope for comparison, anchored arbitrarily at M 2.5.

5.6 Sensitivity experiments

Two variants were evaluated on the same folds. *Tail weighting*, which multiplies record weights by the inverse frequency of the magnitude bin (capped at 10), reduced the underestimation of large events only modestly (-0.57 instead of -0.67 at M 4.0–4.5; MAE on M_w events 0.609 instead of 0.687) while increasing the overall MAE from 0.343 to 0.367. Sample imbalance alone therefore does not explain the bias. *Restricting training and testing to M_L events* removed 246 M_w events (median M 4.0). On the same M_L events its MAE (0.302 ± 0.016) did not differ meaningfully from the default model (0.314 ± 0.015), while its underestimation above M 3 increased because fewer large events remained for training. Mixing the two scales is therefore not the main limitation.

6 Discussion and conclusions

Amplitude-independent, SNR-insensitive descriptors still carry substantial information on earthquake size, and a data-driven ranking concentrates it in the quantity predicted by source theory: the attenuation-corrected corner frequency aggregated over the network. The within-event audit was essential to this interpretation, since several apparently relevant descriptors (notably high-frequency spectral and complexity measures) were found to track station SNR.

The main limitation is the compressed scaling of the apparent corner frequency and the resulting underestimation of larger events. Plausible causes, not yet separated, are:

- window length: the 2 s P and 5 s S windows limit spectral resolution (0.5 and 0.2 Hz) and the observable source duration, affecting larger events most;
- the fixed regional $Q(f)$ and the uncorrected site κ ;
- the analysis band (0.7 or 1 to 40 Hz), which truncates high corners.

Further limitations are the mixed M_L/M_w labels, whole-second origin times, the catalogue limit at M 2.0 (which causes the low-end bias), residual sub-threshold SNR sensitivity, untested sensitivity to λ , $Q(f)$ and window lengths, and a single network over about 16 months.

Planned work: longer and multiple windows to test whether the large-magnitude saturation is a resolution effect; sensitivity analyses of $Q(f)$, site κ , the leakage threshold and window lengths, with region- and sequence-held-out validation; a second network and period; an early-warning variant using the first seconds of P; and a full literature review.

Data and code availability

The processing and analysis code, including automated tests and complete feature definitions, is maintained in the project repository. Waveforms are available from the KOERI network (KOERI, 1971) and hypocentres from AFAD (AFAD, 2026).

7 Declaration of AI use

Generative AI was used in this work. The AI system was Claude Opus 5.5 (Anthropic), accessed through the Claude Code command-line agent, in October 2026. It was used for:

1. **Software:** implementation of the data-processing, feature-extraction, event-selection, training and evaluation code and its automated tests, at the author’s direction;
2. **Analysis:** execution of the experiments, the leakage audit and the importance analyses, and production of figures and tables;

3. **Literature:** identification of candidate references by web search and verification of their bibliographic data against Crossref;
4. **Writing:** drafting of this manuscript.

The research question and the main methodological decisions were set by the author. These decisions include requiring amplitude independence, adding physics-based descriptors, excluding SNR-sensitive descriptors by default, balancing magnitudes and reporting only leakage-filtered results. All numerical results were produced by the code in the project repository and can be regenerated from it. The author has reviewed this AI-drafted preliminary report and takes responsibility for its content.

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